Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim : Design Butterworth and Chebyshev filter using bilinear transformation method.**	
Experiment : 4	**Date: 9/10/2026**	**Enrolment No:92400133050**

AIM: Design Butterworth and Chebyshev filter using bilinear transformation method.

Theory: The bilinear transformation method is commonly used to design analog filters and then convert them into digital filters. This method maps the analog frequency response to the digital frequency response using a bilinear transformation.

The Butterworth and Chebyshev filters are two commonly used filter types. The Butterworth filter has a maximally flat frequency response in the passband, while the Chebyshev filter allows for a sharper transition between the passband and the stopband at the expense of ripples in either the passband or stopband.

Code :

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import butter, bilinear, freqz, cheby1

def design_butterworth_filter(filter_order, cutoff_frequency, sampling_frequency):
    # Design the analog Butterworth filter
    analog_b, analog_a = butter(filter_order, cutoff_frequency, analog=True, btype='low')
    # Perform the bilinear transformation
    digital_b, digital_a = bilinear(analog_b, analog_a, sampling_frequency)
    return digital_b, digital_a

def design_chebyshev_filter(filter_order, cutoff_frequency, sampling_frequency, ripple):
    # Design the analog Chebyshev filter
    analog_b, analog_a = cheby1(filter_order, ripple, cutoff_frequency, analog=True, btype='low')
    # Perform the bilinear transformation
    digital_b, digital_a = bilinear(analog_b, analog_a, sampling_frequency)
    return digital_b, digital_a

def plot_filter_response(digital_b, digital_a, sampling_frequency):
    # Compute the frequency response of the filter
    frequency, magnitude_response = freqz(digital_b, digital_a, fs=sampling_frequency)
    # Plot the magnitude response
    plt.figure(figsize=(10, 6))
    plt.plot(frequency, np.abs(magnitude_response))
    plt.title('Filter Magnitude Response')
    plt.xlabel('Frequency (Hz)')
    plt.ylabel('Magnitude')
    plt.grid(True)
    plt.show()

    # Compute the impulse response of the filter
    _, impulse_response = freqz(digital_b, digital_a, fs=sampling_frequency, worN=4096)
    # Plot the impulse response
    plt.figure(figsize=(10, 6))
    plt.plot(impulse_response)
    plt.title('Filter Impulse Response')
    plt.xlabel('Samples')
    plt.ylabel('Amplitude')
    plt.grid(True)
    plt.show()

# Specify the desired filter specifications
filter_order = 4 # Filter order
cutoff_frequency = 1000 # Cutoff frequency in Hz
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim : Design Butterworth and Chebyshev filter using bilinear transformation method.**	
Experiment : 4	**Date: 9/10/2026**	**Enrolment No:92400133050**

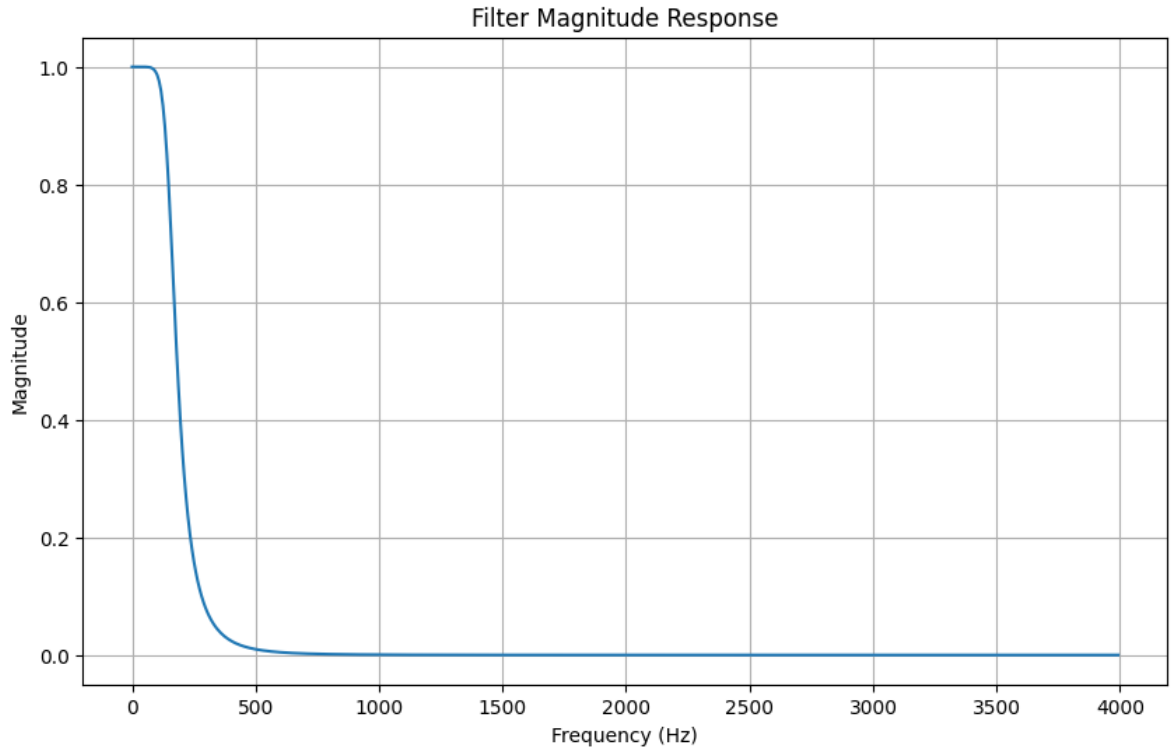
```
sampling_frequency = 8000 # Sampling frequency in Hz
ripple = 0.5 # Ripple factor for Chebyshev filter

# Design the Butterworth filter
digital_b, digital_a = design_butterworth_filter(filter_order, cutoff_frequency, sampling_frequency)
# Plot the Butterworth filter's magnitude response and impulse response
plot_filter_response(digital_b, digital_a, sampling_frequency)

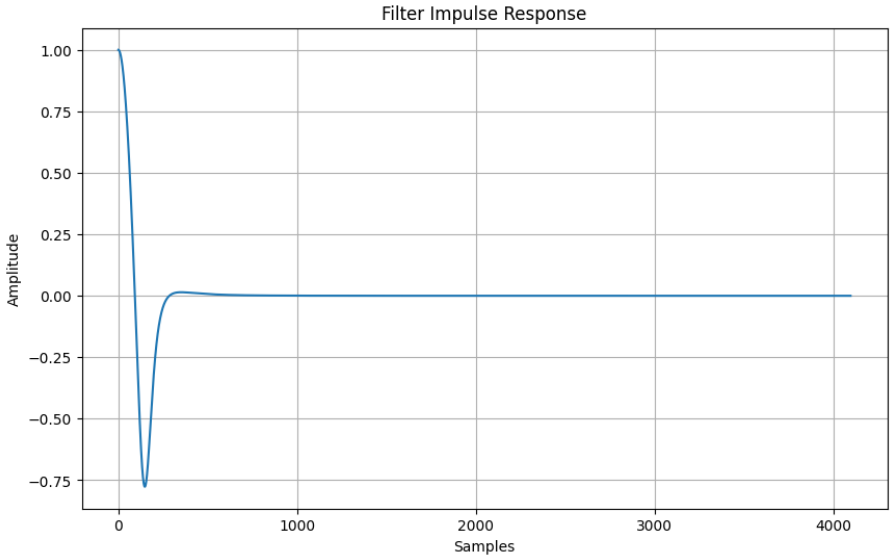
# Design the Chebyshev filter
digital_b, digital_a = design_chebyshev_filter(filter_order, cutoff_frequency, sampling_frequency, ripple)
# Plot the Chebyshev filter's magnitude response and impulse response
plot_filter_response(digital_b, digital_a, sampling_frequency)

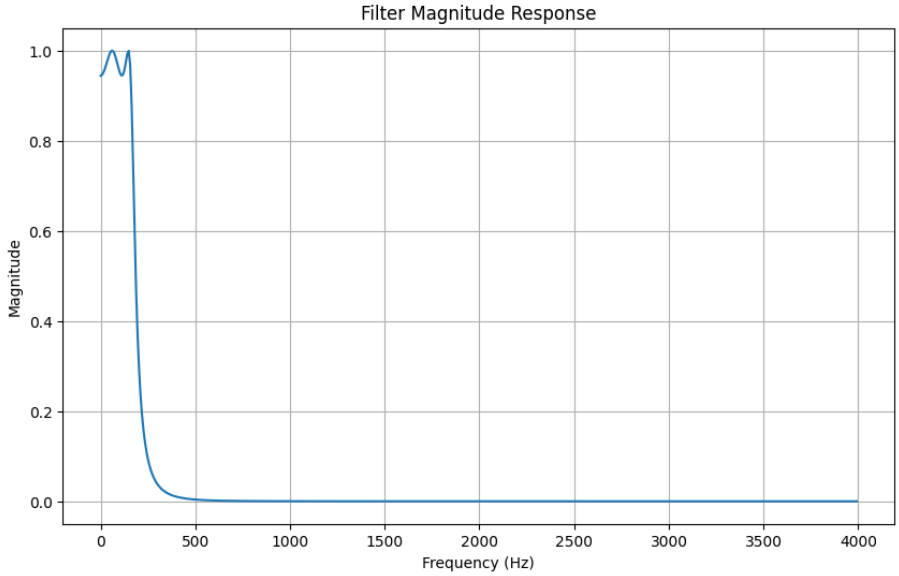
# Save the filter coefficients (optional)
filter_path = 'filter_coefficients.txt'
np.savetxt(filter_path, np.vstack((digital_b, digital_a)), delimiter=',')
print(f"Filter coefficients saved at: {filter_path}")
```

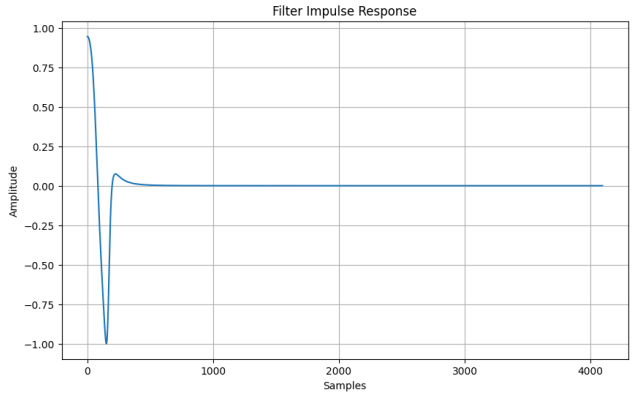
Output :



Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim : Design Butterworth and Chebyshev filter using bilinear transformation method.**	
Experiment : 4	**Date: 9/10/2026**	**Enrolment No:92400133050**







Filter coefficients saved at: filter_coefficients.txt

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim : Design Butterworth and Chebyshev filter using bilinear transformation method.**	
Experiment : 4	**Date: 9/10/2026**	**Enrolment No:92400133050**

Additional Task

•Generate a discrete sine wave of 1KHz, 2KHz, 3KHz, 4 KHz. Plot all these

Code :

```
import numpy as np
import matplotlib.pyplot as plt

fs = 10000
t = np.arange(0, 0.01, 1/fs)

x1 = np.sin(2*np.pi*1000*t)
x2 = np.sin(2*np.pi*2000*t)
x3 = np.sin(2*np.pi*3000*t)
x4 = np.sin(2*np.pi*4000*t)

plt.figure(figsize=(10,8))

plt.subplot(4,1,1)
plt.stem(t,x1)
plt.title("1 kHz Sine Wave")
plt.grid()

plt.subplot(4,1,2)
plt.stem(t,x2)
plt.title("2 kHz Sine Wave")
plt.grid()

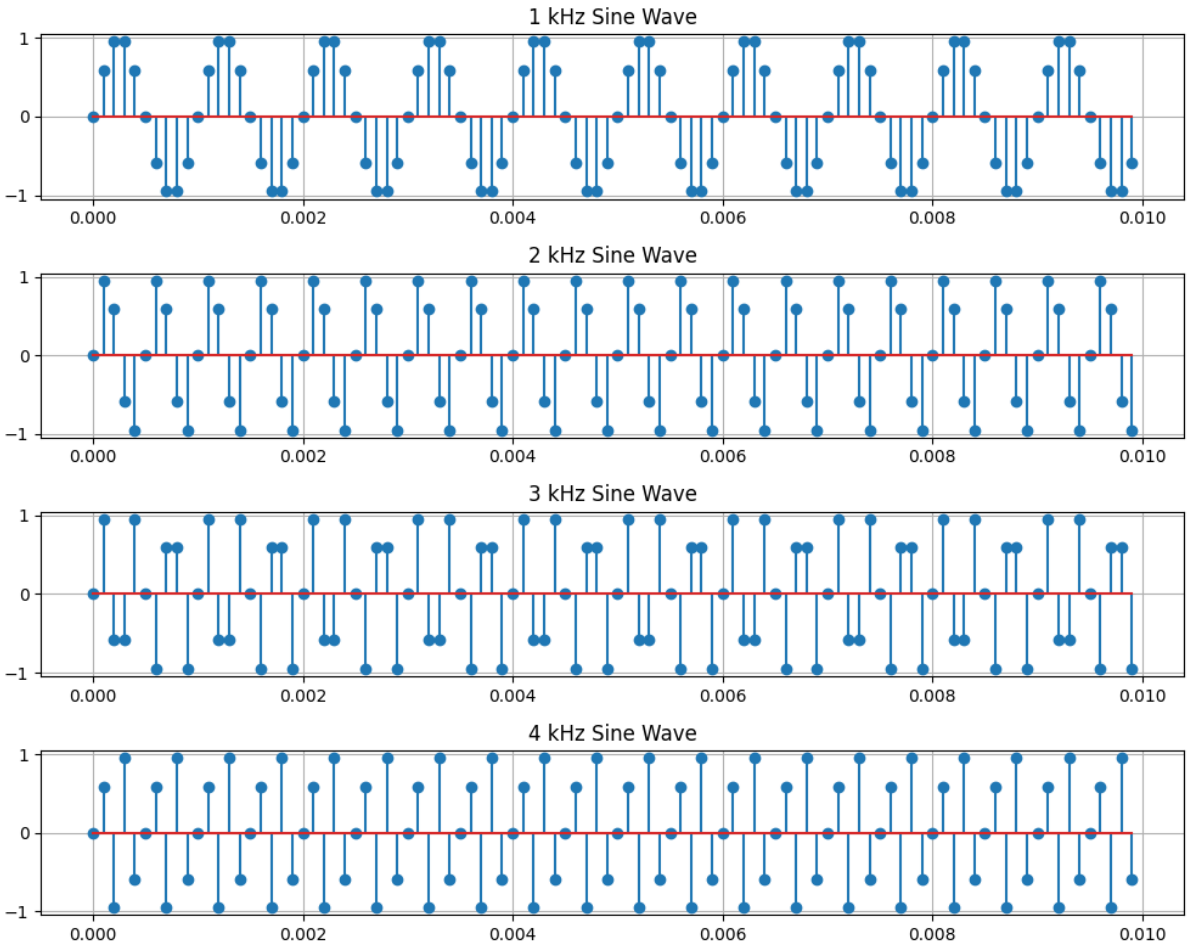
plt.subplot(4,1,3)
plt.stem(t,x3)
plt.title("3 kHz Sine Wave")
plt.grid()

plt.subplot(4,1,4)
plt.stem(t,x4)
plt.title("4 kHz Sine Wave")
plt.grid()

plt.tight_layout()
plt.show()
```

Output :

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim : Design Butterworth and Chebyshev filter using bilinear transformation method.**	
Experiment : 4	**Date: 9/10/2026**	**Enrolment No:92400133050**



•Add all these four signal and plot it in time as well as frequency domain

Code :

```
import numpy as np
import matplotlib.pyplot as plt

fs = 10000
t = np.arange(0, 0.01, 1/fs)

x1 = np.sin(2*np.pi*1000*t)
x2 = np.sin(2*np.pi*2000*t)
x3 = np.sin(2*np.pi*3000*t)
x4 = np.sin(2*np.pi*4000*t)

x = x1 + x2 + x3 + x4

# Time domain
plt.figure(figsize=(10,4))
plt.stem(t,x)
plt.title("Added Signal - Time Domain")
plt.xlabel("Time (s)")
```

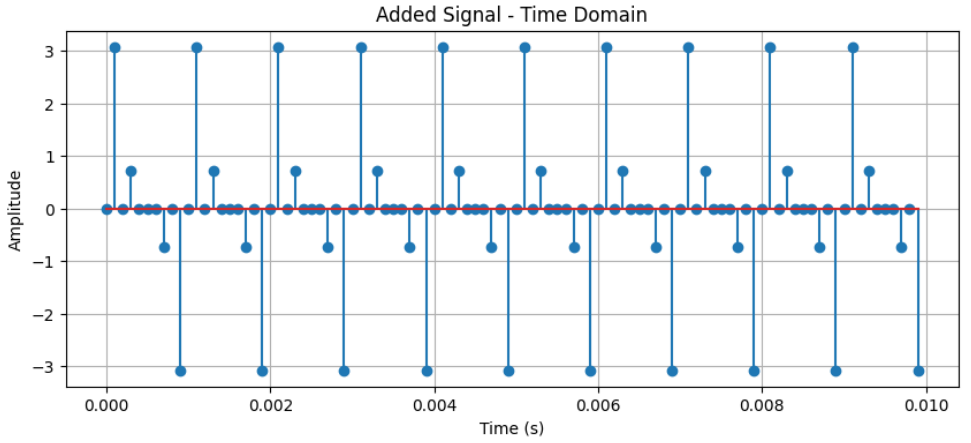
Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim : Design Butterworth and Chebyshev filter using bilinear transformation method.**	
Experiment : 4	**Date: 9/10/2026**	**Enrolment No:92400133050**

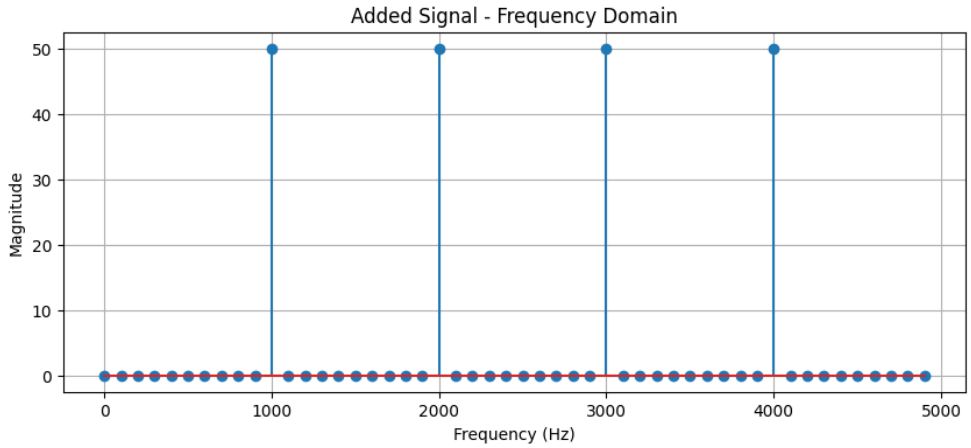
```
plt.ylabel("Amplitude")
plt.grid()
plt.show()

# Frequency domain
X = np.fft.fft(x)
freq = np.fft.fftfreq(len(x),1/fs)

plt.figure(figsize=(10,4))
plt.stem(freq[:len(freq)//2],np.abs(X[:len(X)//2]))
plt.title("Added Signal - Frequency Domain")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid()
plt.show()
```

Output :





Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim : Design Butterworth and Chebyshev filter using bilinear transformation method.**	
Experiment : 4	**Date: 9/10/2026**	**Enrolment No:92400133050**

• pass through buttorworth digital filter having cutoff frequency 2 khz and plot the output plot in time as well as frequency domain

Code :

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import butter, filtfilt

fs = 10000
t = np.arange(0, 0.01, 1/fs)

x1 = np.sin(2*np.pi*1000*t)
x2 = np.sin(2*np.pi*2000*t)
x3 = np.sin(2*np.pi*3000*t)
x4 = np.sin(2*np.pi*4000*t)

x = x1 + x2 + x3 + x4

# Butterworth filter
b,a = butter(4,2000/(fs/2),btype='low')

y = filtfilt(b,a,x)

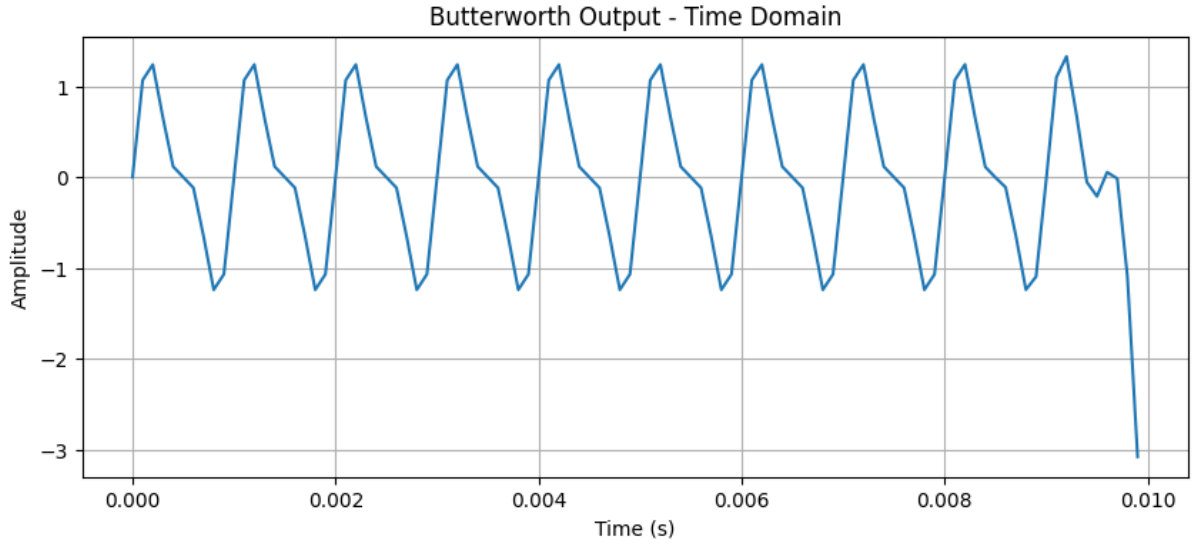
# Time domain
plt.figure(figsize=(10,4))
plt.plot(t,y)
plt.title("Butterworth Output - Time Domain")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid()
plt.show()

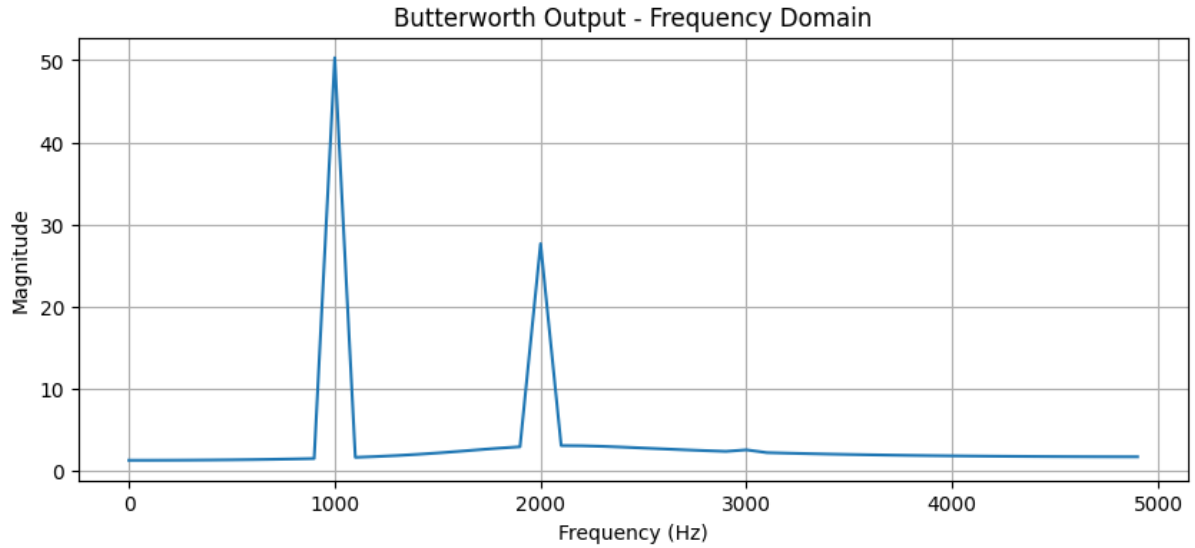
# Frequency domain
Y = np.fft.fft(y)
freq = np.fft.fftfreq(len(y),1/fs)

plt.figure(figsize=(10,4))
plt.plot(freq[:len(freq)//2],np.abs(Y[:len(Y)//2]))
plt.title("Butterworth Output - Frequency Domain")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid()
plt.show()
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim : Design Butterworth and Chebyshev filter using bilinear transformation method.**	
Experiment : 4	**Date: 9/10/2026**	**Enrolment No:92400133050**

Output :





•Pass through chebyschev digital filter having cutoff frequency 3KHz and plot the output plot in time as well as frequency domain

Code :

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import cheby1, filtfilt

fs = 10000
t = np.arange(0, 0.01, 1/fs)
```

Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim : Design Butterworth and Chebyshev filter using bilinear transformation method.**	
Experiment : 4	**Date: 9/10/2026**	**Enrolment No:92400133050**

```
x1 = np.sin(2*np.pi*1000*t)
x2 = np.sin(2*np.pi*2000*t)
x3 = np.sin(2*np.pi*3000*t)
x4 = np.sin(2*np.pi*4000*t)

x = x1 + x2 + x3 + x4

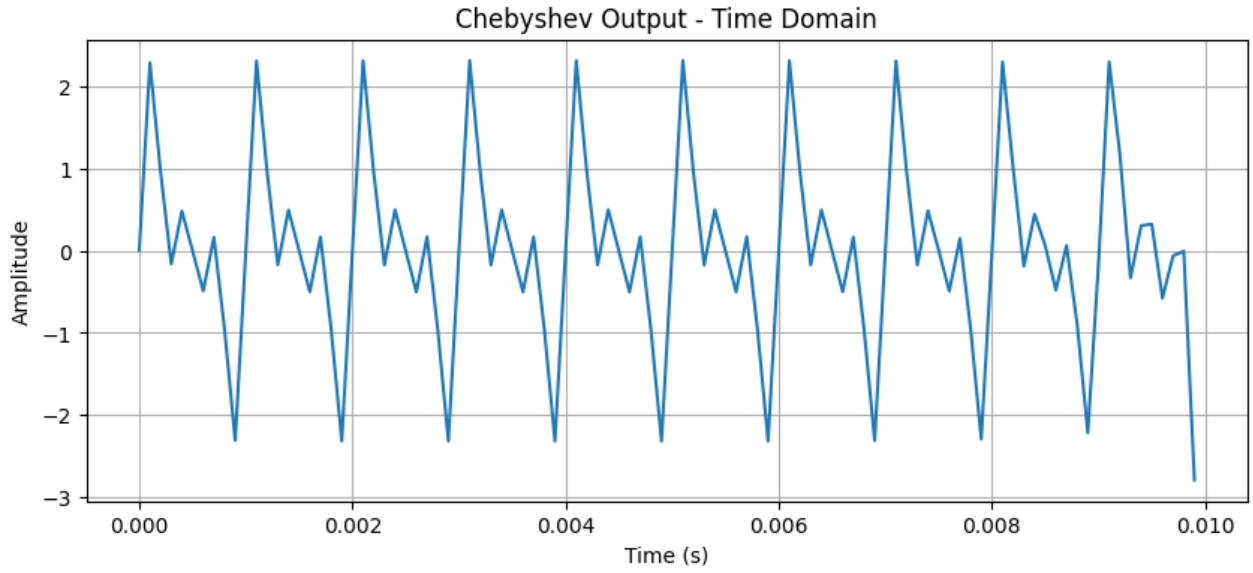
# Chebyshev filter
b, a = cheby1(4, 0.5, 3000/(fs/2), btype='low')
y = filtfilt(b, a, x)

# Time domain
plt.figure(figsize=(10, 4))
plt.plot(t, y)
plt.title("Chebyshev Output - Time Domain")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid()
plt.show()

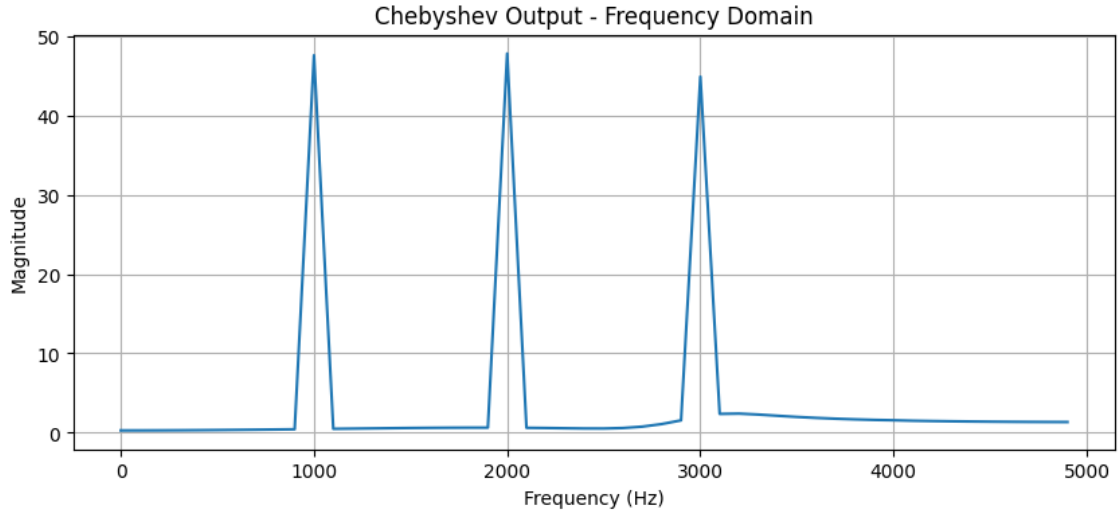
# Frequency domain
Y = np.fft.fft(y)
freq = np.fft.fftfreq(len(y), 1/fs)

plt.figure(figsize=(10, 4))
plt.plot(freq[:len(freq)//2], np.abs(Y[:len(Y)//2]))
plt.title("Chebyshev Output - Frequency Domain")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid()
plt.show()
```

Output :



Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim : Design Butterworth and Chebyshev filter using bilinear transformation method.**	
Experiment : 4	**Date: 9/10/2026**	**Enrolment No:92400133050**



Conclusion :

In this experiment, we created digital low-pass filters using Butterworth and Chebyshev designs. We generated four sine waves with different frequencies and combined them into one signal. After passing the signal through each filter, the higher frequencies were reduced according to the cutoff. The frequency plots showed how each filter changes the signal. This helped us understand the smooth Butterworth response and the rippled Chebyshev response.